

GREYBULL, WY, USA

WMO: 726667

Lat: 44.517N

Lon: 108.082W

Elev: 3933

StdP: 12.72

Time Zone: -7.00 (NAM)

Period: 99-19

WBAN: 24048

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.3	-7.3	-18.1	2.4	-12.8	-11.8	3.4	-5.8	31.0	25.2	26.2	25.5	2.3	130	0.606

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.4	97.3	62.2	94.4	61.3	91.2	60.6	65.6	88.2	63.7	86.0	62.2	85.0	8.2	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.8	82.5	71.7	55.5	76.0	70.1	54.0	71.7	69.3	32.5	86.6	30.8	85.7	29.7	84.9	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
28.3	24.4	20.1	DB	-23.1	102.3	6.3	2.8	-27.7	104.3	-31.4	105.9	-34.9	107.5	-39.5	109.5	
			WB	-23.4	67.4	6.2	3.6	-27.9	70.0	-31.5	72.2	-35.0	74.2	-39.5	76.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	46.4	17.5	22.5	38.0	46.8	56.5	67.0	75.9	72.2	61.3	46.8	32.0	18.7	(6)
(7)		DBStd	21.87	10.94	12.24	10.10	8.15	8.30	7.38	4.94	5.55	7.73	8.62	11.51	10.72	(7)
(8)		HDD50	4023	1008	770	380	152	32	1	0	0	9	159	542	970	(8)
(9)		HDD65	7554	1473	1190	836	545	285	61	2	13	156	567	991	1435	(9)
(10)		CDD50	2713	0	0	9	58	233	513	804	689	347	59	1	0	(10)
(11)		CDD65	767	0	0	0	1	20	123	341	236	45	1	0	0	(11)
(12)		CDH74	11462	0	0	4	53	447	1831	4749	3314	1005	59	0	0	(12)
(13)		CDH80	5641	0	0	0	7	130	823	2607	1671	393	10	0	0	(13)
(14)	Wind	WSAvg	6.7	4.3	5.6	7.6	9.2	8.6	8.0	7.1	6.6	6.4	6.5	5.5	4.9	(14)
(15)	Precipitation	PrecAvg	7.0	0.2	0.2	0.4	0.7	1.2	1.3	0.7	0.4	0.7	0.8	0.3	0.2	(15)
(16)		PrecMax	9.9	0.8	0.7	1.0	2.2	2.6	3.2	2.7	1.3	1.9	1.8	0.8	0.7	(16)
(17)		PrecMin	4.9	0.0	0.0	0.0	0.2	0.3	0.4	0.0	0.1	0.1	0.2	0.0	0.0	(17)
(18)		PrecStd	1.5	0.2	0.2	0.2	0.4	0.7	0.8	0.7	0.3	0.5	0.5	0.2	0.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.9	55.7	72.1	79.9	88.7	98.6	101.7	99.3	93.3	80.6	64.6	48.1	(19)
(20)		MCWB	37.6	41.3	48.9	52.4	57.9	61.2	63.3	64.4	59.8	54.9	46.9	38.6		(20)
(21)		2%	DB	40.8	49.3	65.7	73.9	83.7	93.3	98.9	95.6	89.2	73.8	58.5	42.5	(21)
(22)		MCWB	33.6	38.6	46.0	50.0	56.1	61.0	64.2	60.8	58.6	52.1	43.6	34.8		(22)
(23)		5%	DB	36.6	44.2	60.8	69.2	79.4	89.4	95.9	92.7	84.4	68.6	53.5	37.3	(23)
(24)		MCWB	31.3	35.3	43.9	48.7	55.3	59.7	62.0	60.6	56.4	50.4	41.1	31.5		(24)
(25)		10%	DB	33.1	38.9	55.1	63.9	74.0	84.4	92.7	89.8	80.4	63.1	47.9	34.0	(25)
(26)		MCWB	29.0	32.5	41.5	46.3	53.0	58.6	61.5	60.0	55.6	47.8	38.7	29.6		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.0	43.2	50.2	54.1	60.2	65.4	70.2	68.2	62.7	56.4	47.8	39.6	(27)
(28)		MCDWB	45.5	53.5	69.4	76.5	81.6	83.5	97.8	94.2	82.0	75.6	62.4	46.7		(28)
(29)		2%	WB	34.3	39.2	46.9	51.1	58.1	62.8	66.5	64.3	60.2	53.6	44.2	35.1	(29)
(30)		MCDWB	40.0	48.4	63.7	69.9	77.8	85.5	90.9	86.9	81.6	70.0	57.1	41.2		(30)
(31)		5%	WB	31.7	35.7	44.5	49.3	56.3	61.2	64.6	62.5	58.5	51.3	41.7	32.2	(31)
(32)		MCDWB	36.3	43.0	59.3	66.8	74.8	84.5	87.3	84.8	79.8	66.3	51.9	37.3		(32)
(33)		10%	WB	29.0	32.8	42.1	47.2	54.4	59.6	63.1	61.2	56.4	48.9	39.0	29.6	(33)
(34)		MCDWB	33.0	38.9	54.4	63.4	71.1	81.0	85.7	83.5	76.6	61.8	47.2	33.8		(34)
(35)	Mean Daily Temperature Range	MDBR	22.3	22.4	25.6	25.4	25.0	27.8	31.4	31.3	30.1	25.4	23.9	21.2	(35)	
(36)		5% DB	MCDBR	24.8	26.9	34.7	35.1	33.3	35.3	36.2	36.7	38.9	35.7	31.4	24.2	(36)
(37)		MCWBR	18.6	17.3	18.6	16.4	12.7	11.7	12.1	13.1	15.3	17.8	18.8	17.9	(37)	
(38)		5% WB	MCDBR	24.2	25.9	31.9	32.1	30.4	32.2	32.3	31.8	33.8	32.8	29.3	23.6	(38)
(39)		MCWBR	18.4	17.2	17.5	15.2	11.7	11.2	12.2	12.2	13.7	16.9	17.9	17.7	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.249	0.254	0.279	0.317	0.339	0.344	0.363	0.358	0.325	0.286	0.260	0.245	(40)	
(41)		taud	2.523	2.525	2.480	2.369	2.374	2.386	2.365	2.359	2.442	2.530	2.532	2.527	(41)	
(42)		Ebn at Noon	286	304	307	300	295	292	285	283	285	286	279	277	(42)	
(43)		Edh at Noon	21	25	30	36	37	37	37	36	31	24	20	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	566	875	1288	1638	1916	2263	2294	1951	1534	992	636	459	(44)	
(45)		RadStd	41	56	89	81	175	148	89	84	106	78	53	27	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page